ELEKTRİK DEVRE TEMELLERİ DERS NOTLARI

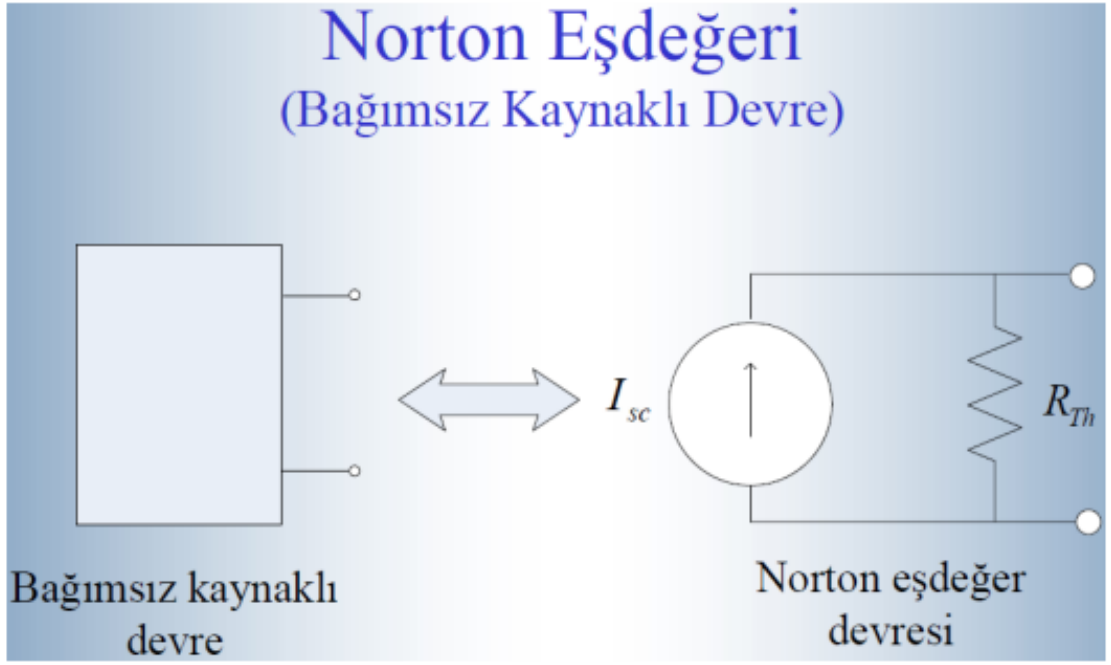
8. HAFTA

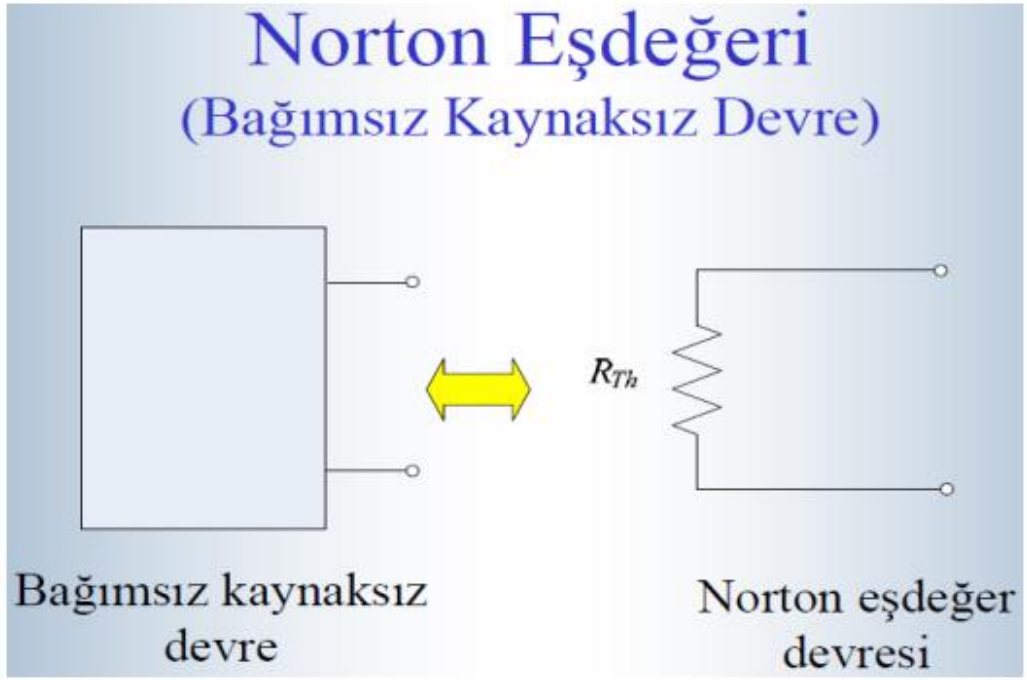
Norton Teoremi ve Maksimum Güç Aktarımı

NORTON TEOREMLERİ

- Her Thevenin eşdeğer devresi, kendisine paralel bir direnç bağlı akım kaynağına dönüştürülebilir (Kaynak dönüşümü).
- Kendisine paralel bir direnç bağlı akım kaynağı Norton eşdeğer devresi olarak adlandırılır.
- Norton eşdeğer devresini bulmak esasen Thevenin eşdeğer devresini bulmak ile aynıdır.

NORTON TEOREMLERİ





Norton Analizi İşlem Basamakları

1.Kısa devre akımını hesapla, I_{SC}

2.Thevenin/Norton eşdeğer direncini hesapla, R_N

(a) Eğer sadece bağımsız kaynak varsa bütün gerilim kaynakları kısa devre, bütün akım kaynakları açık devre yapılır.

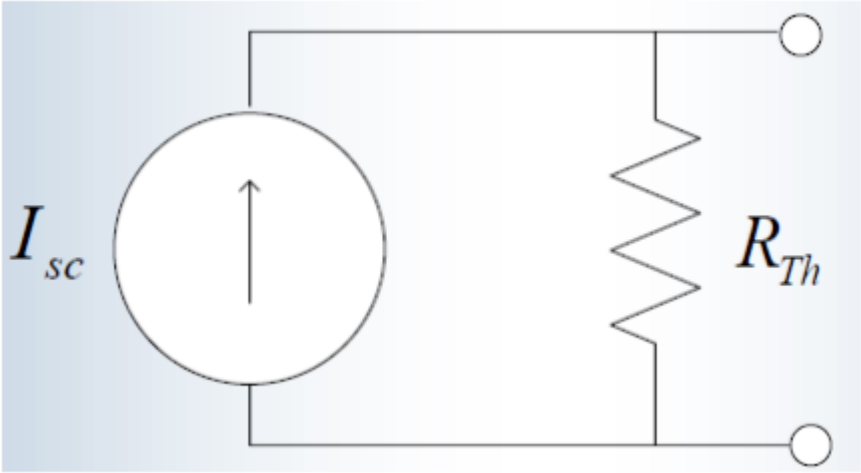
(b) Eğer sadece bağımlı kaynak varsa hesaplama için bir test gerilim(akım) kaynağı kullanılır.

$$R_{Th} = V_{Test} / I_{test}$$

(c) hem bağımlı hem de bağımsız kaynak birlikte kullanılmışsa V_{OC}/I_{SC} den R_{Th} hesaplanır.

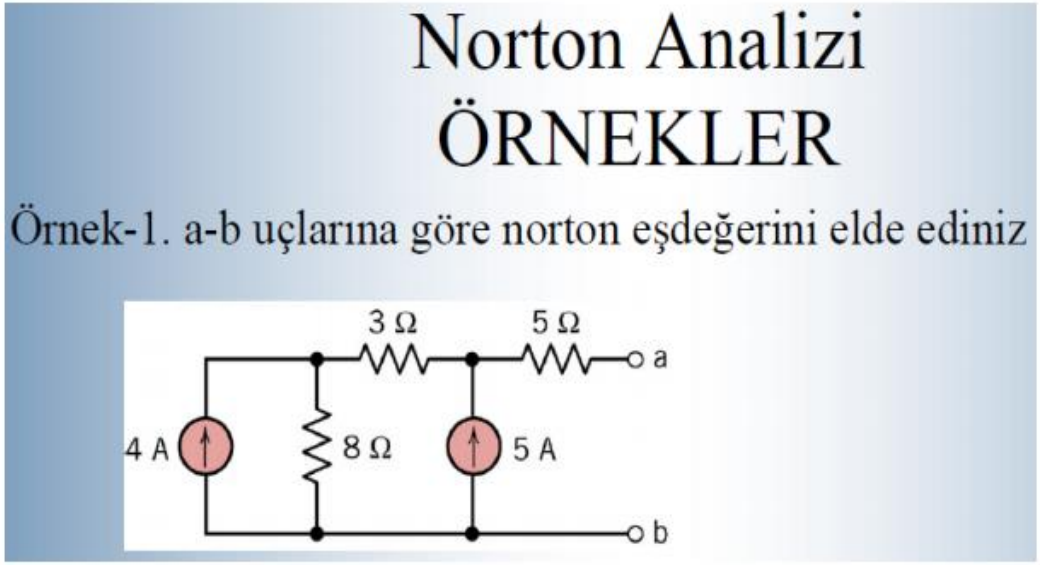
Norton Analizi İşlem Basamakları

3. Devre I_{SC} ve ona paralel bağlı R_{Th} / R_N dönüştürülür.

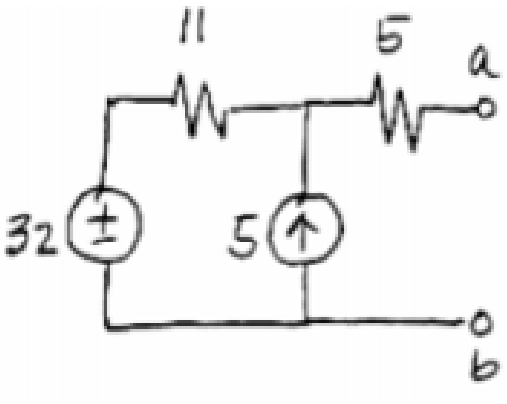


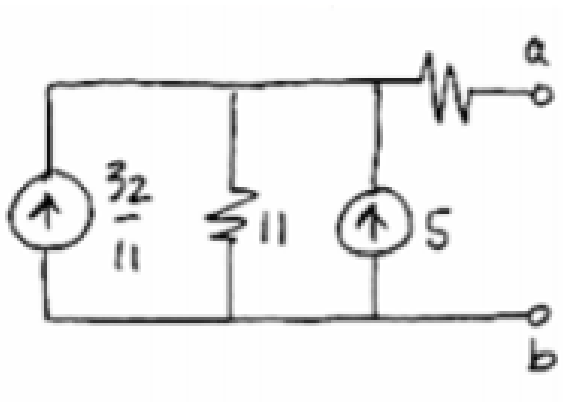
Not: Adım 2(b)de eş değer devre sadece R_{Th} den oluşur, akım kaynağı yoktur.

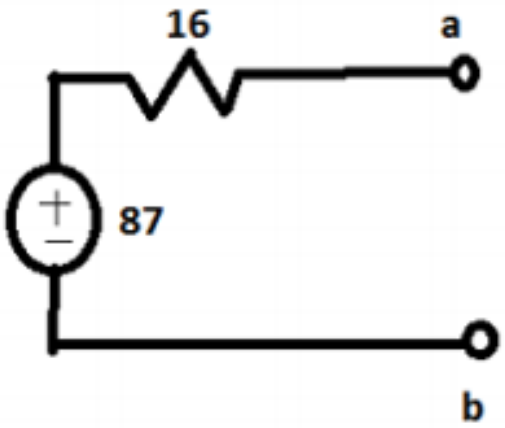
Norton Analizi Örnekler

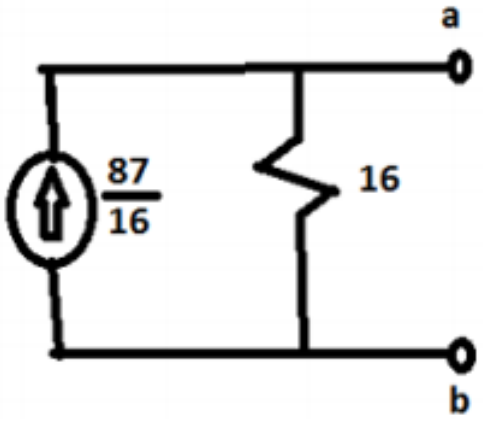


- Kaynak dönüşümüyle yapabilir miyiz?
- Bakalım



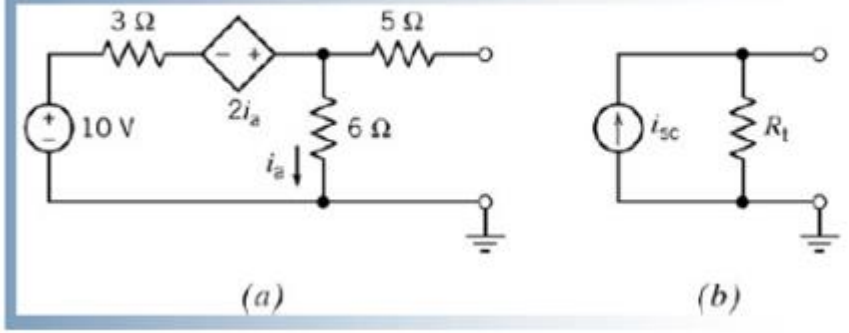


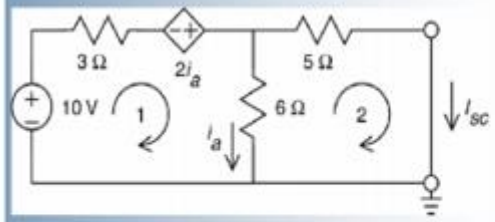




Norton Analizi Örnekler

Örnek-1. Norton eşdeğerini bulunuz





$$i_a = i_1 - i_2 = i_1 - i_{sc}$$

$$3i_1 - 2(i_1 - i_2) + 6(i_1 - i_2) - 10 = 0 \Rightarrow 7i_1 - 4i_2 = 10$$

$$5i_2 - 6(i_1 - i_2) = 0 \Rightarrow -6i_1 + 11i_2 = 0 \Rightarrow i_1 = \frac{11}{6} i_2$$

$$7\left(\frac{11}{6} i_2\right) - 4i_2 = 10 \Rightarrow i_2 = 1.13\text{ A} \Rightarrow i_{sc} = 1.13\text{ A}$$

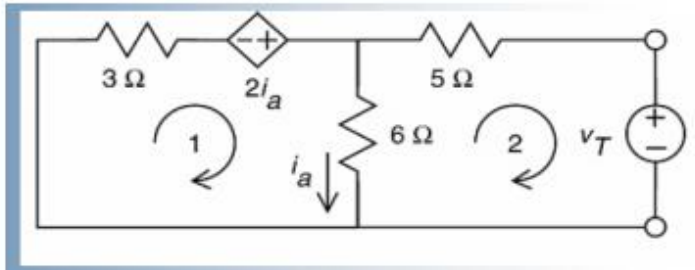
Voc hesaplayarak Rth= Vos/Isc ile bulalım

-10+3Ia-2Ia+6Ia=0
Ia=10/7= 1.42A

Voc=6*Ia=6*1.42=8.57V

Rth=Vos/Isc
=8.57/1.13
=7.57 ohm

Gerilim kaynağını kaldırıp sanal kaynak uygulayarak Rth bulalım



$$i_a = i_1 - i_2 = i_1 + i_T$$

$$3i_1 - 2(i_1 - i_2) + 6(i_1 - i_2) = 0 \Rightarrow 7i_1 - 4i_2 = 0 \Rightarrow i_1 = \frac{4}{7} i_2$$

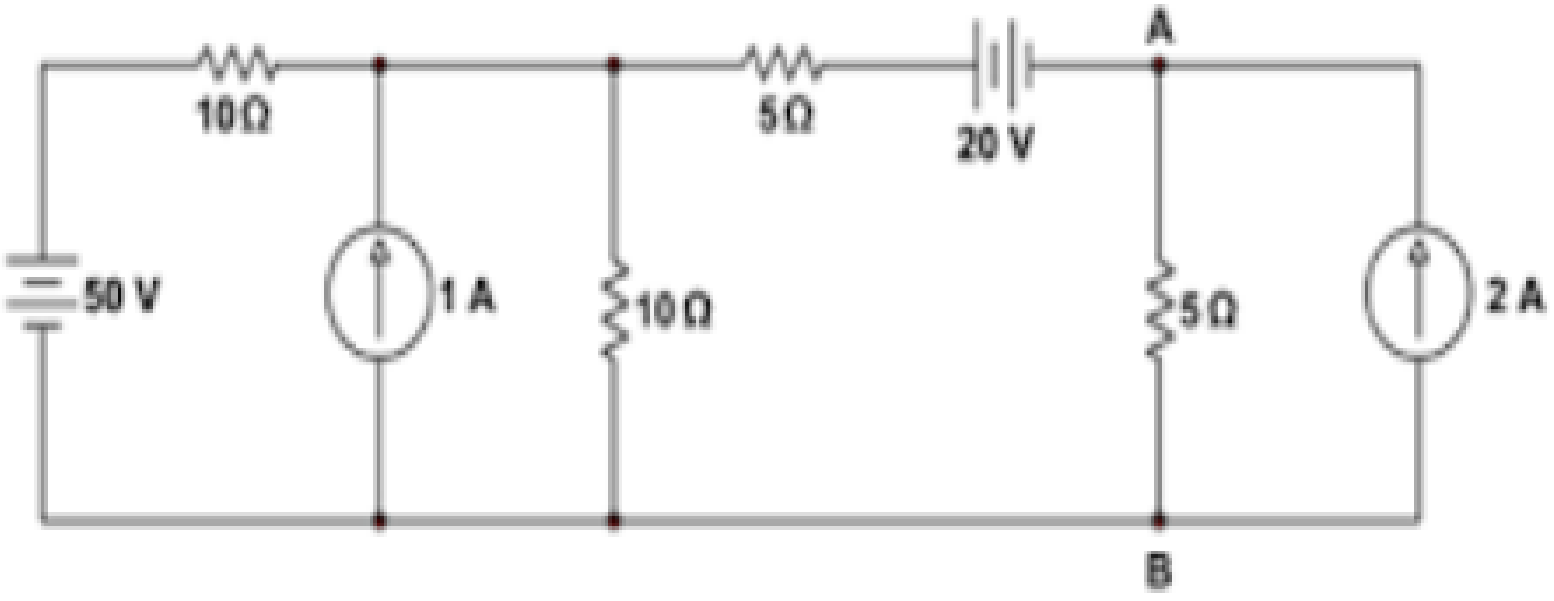
$$5i_2 + v_T - 6(i_1 - i_2) = 0 \Rightarrow -6i_1 + 11i_2 = -v_T$$

$$-6\left(\frac{4}{7} i_2\right) + 11i_2 = -v_T \Rightarrow 7.57\, i_2 = -v_T$$

$$R_t = \frac{v_T}{i_T} = \frac{-v_T}{-i_T} = \frac{-v_T}{i_2} = 7.57\ \Omega$$

Örnek Çözelim

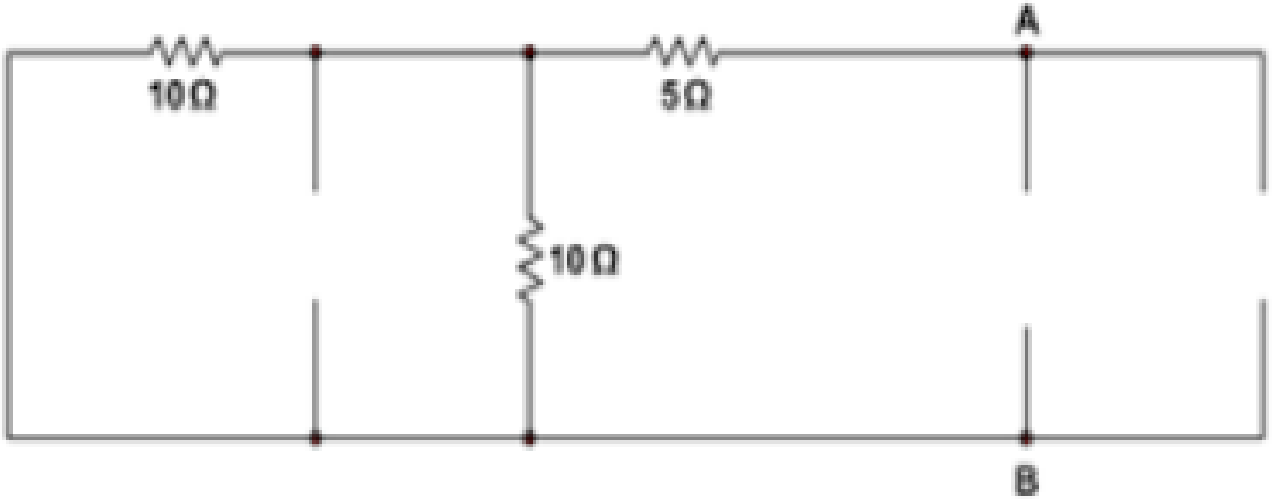
- Aşağıda görüldüğü gibi bir devremiz var. A-B uçları arasını yükümüz kabul edip norton teoremiyle bu devreyi çözelim.



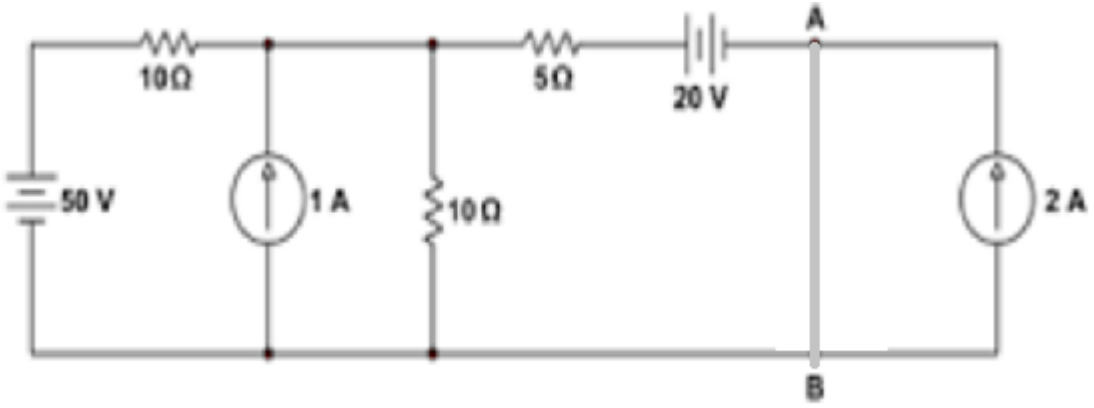
Örnek Çözelim

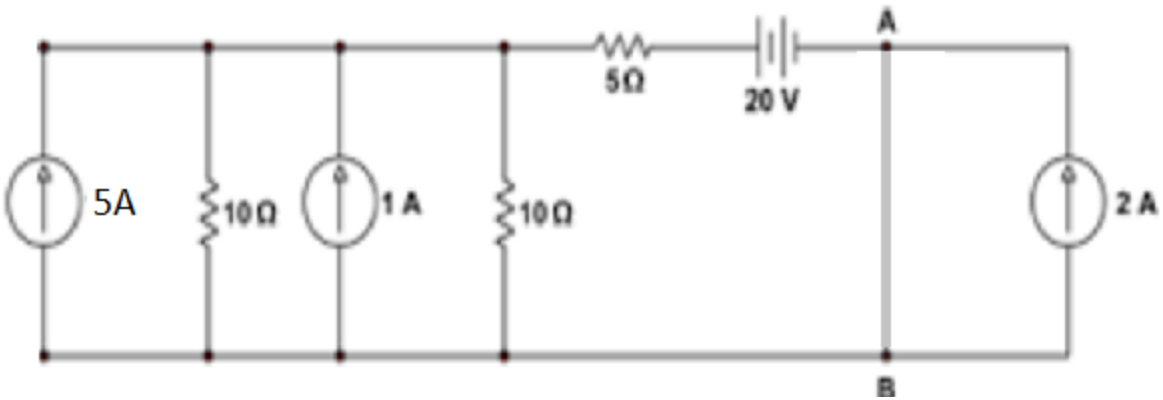
- Rn'yi hesaplayabilmek için tüm kaynaklarımızı devre dışı ediyoruz. Sonuçta aşağıdaki gibi devre karşımıza çıkıyor:

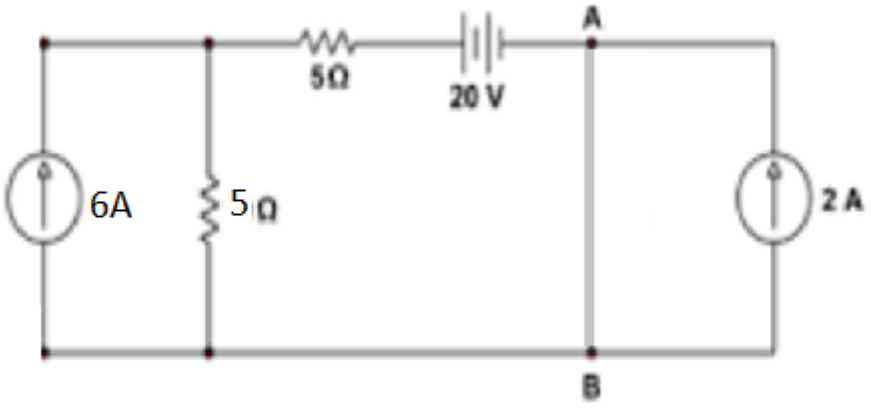
- Rn=10ohm

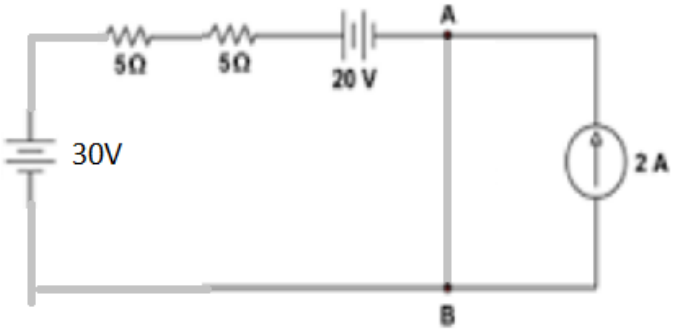


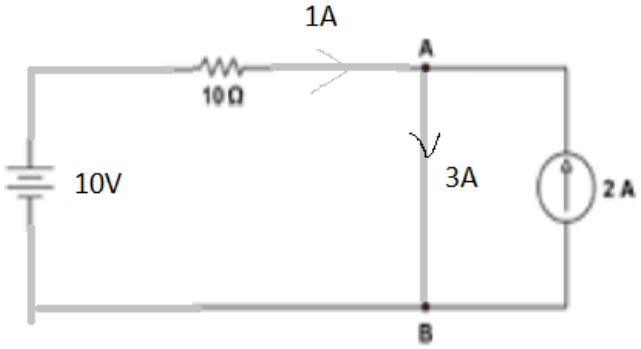
- Kısadevre akımı:

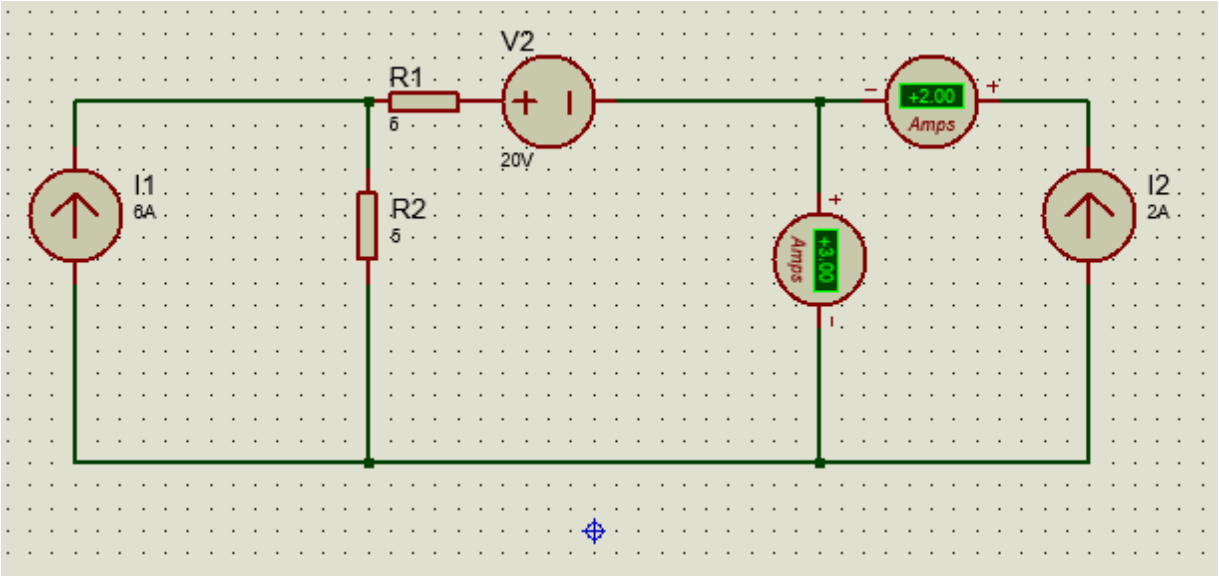




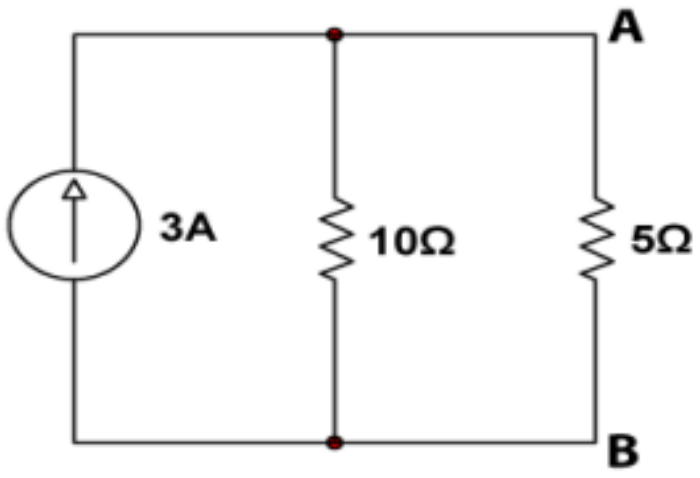



A
5Ω
5Ω
20 V
30V
2 A
B


1A
A
10Ω
10V
3A
2 A
B


V2
R1
6
20V
I1
6A
R2
6
+2.00
Amps
+3.00
Amps
I2
2A

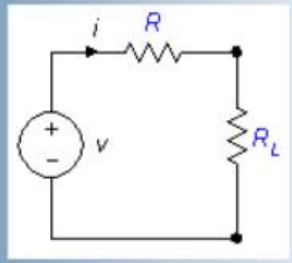
- Sonuçta Rt değeri 10 ohm bulunuyor. Ardından yük direnci devreden çıkartılır ve iki ucu arasındaki akım değeri hesaplanır. Sonuçta devre In = 3A ve A-B uçları arasındaki akım 2A bulunur.



Maksimum Güç Transferi

- Thevenin teoremiyle çalışarak, bir devrenin verebileceği maksimum gücü ve yükün maksimum güç transferi sonucuna uyarlanması yolunu belirleyebiliriz.

EN FAZLA GÜÇ AKTARIMI



Yüke aktarılan güç

$$P_L = i^2 R_L$$

$$= \left(\frac{v}{R+R_L}\right)^2 R_L = \frac{v^2 R_L}{(R+R_L)^2}$$

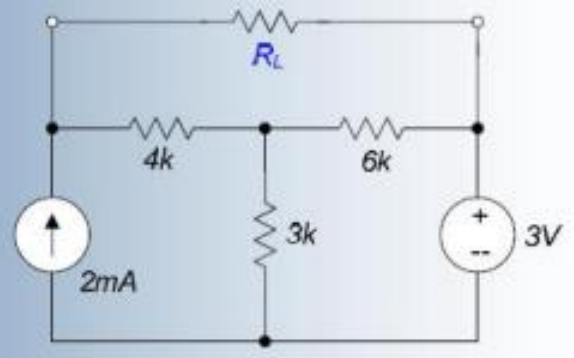
Bu niceliği en yüksek dereceye çıkaran R_L değerini belirleyelim. R_L ile ilgili olarak bu ifadenin türevini alıp sıfıra eşitlersek;

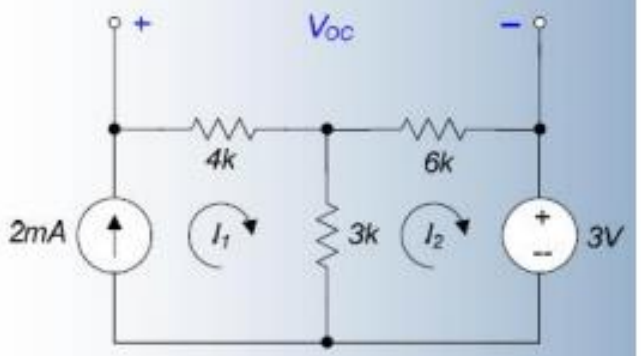
$$\frac{dP_L}{dR_L} = \frac{(R+R_L)^2 v^2 - 2v^2 R_L (R+R_L)}{(R+R_L)^4} = 0$$

$$R_L = R$$

EN FAZLA GÜÇ AKTARIMI

Örnek-1. Aşağıdaki devrede en fazla güç aktarımı için R_L'nin ve bu yüke aktarılabilen en fazla gücün değerini bulunuz.





$$I_1 = 2\times10^{-3}\ \mathbf{A}$$

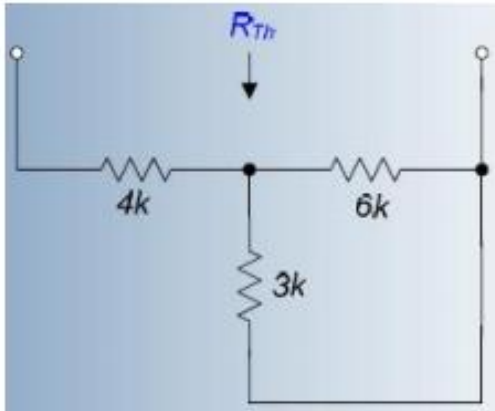
$$3k(I_2 - I_1) + 6kI_2 + 3 = 0$$

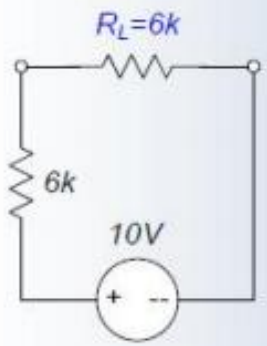
Bu denklemler çözüldüğünde I_2=1/3 mA bulunur ve buradan,

$$V_{OC} = 4kI_1 + 6kI_2$$

$$= 10V$$

EN FAZLA GÜÇ AKTARIMI





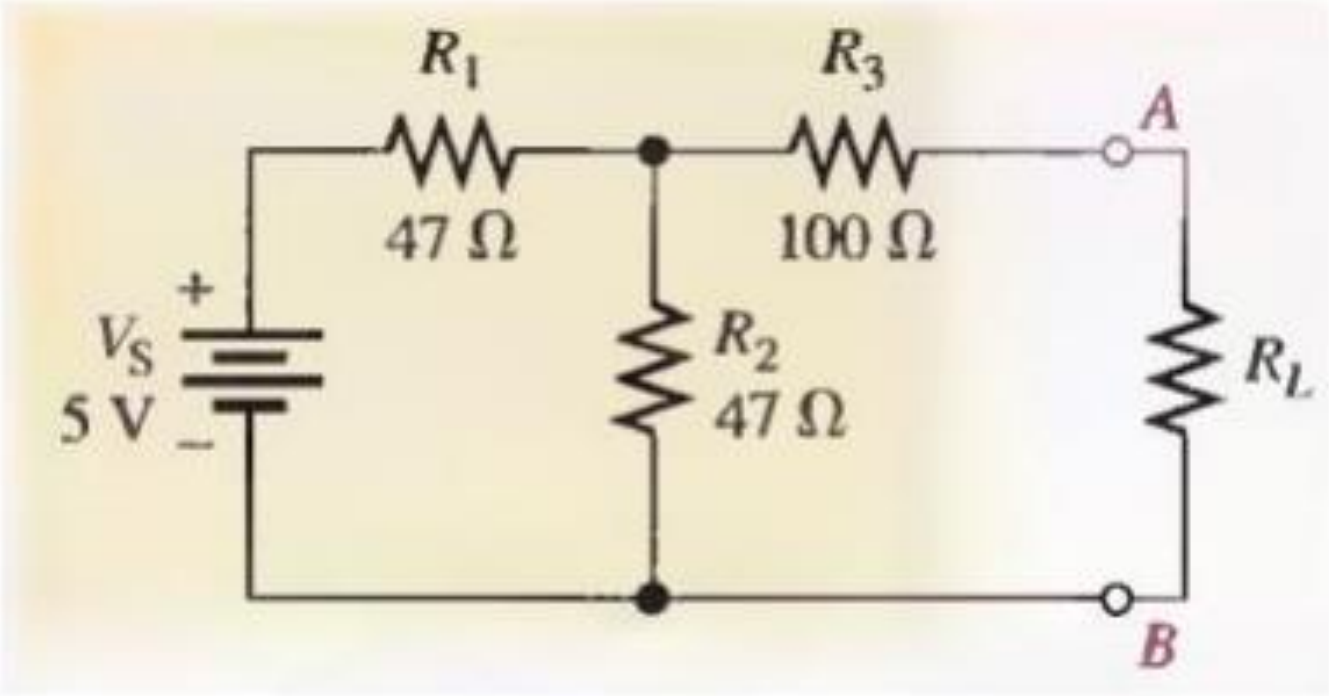
R_{Th} 6kΩ'dur. Buradan, en fazla güç aktarımı için $R_L = R_{Th} = 6k\Omega$ olduğu bulunur.

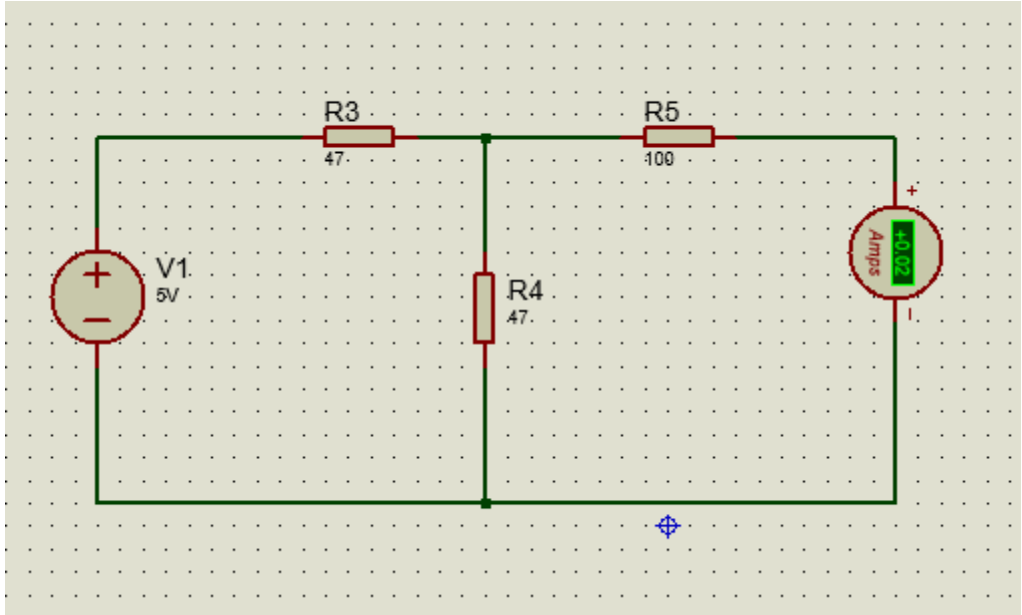
Yüke aktarılan en fazla güç;

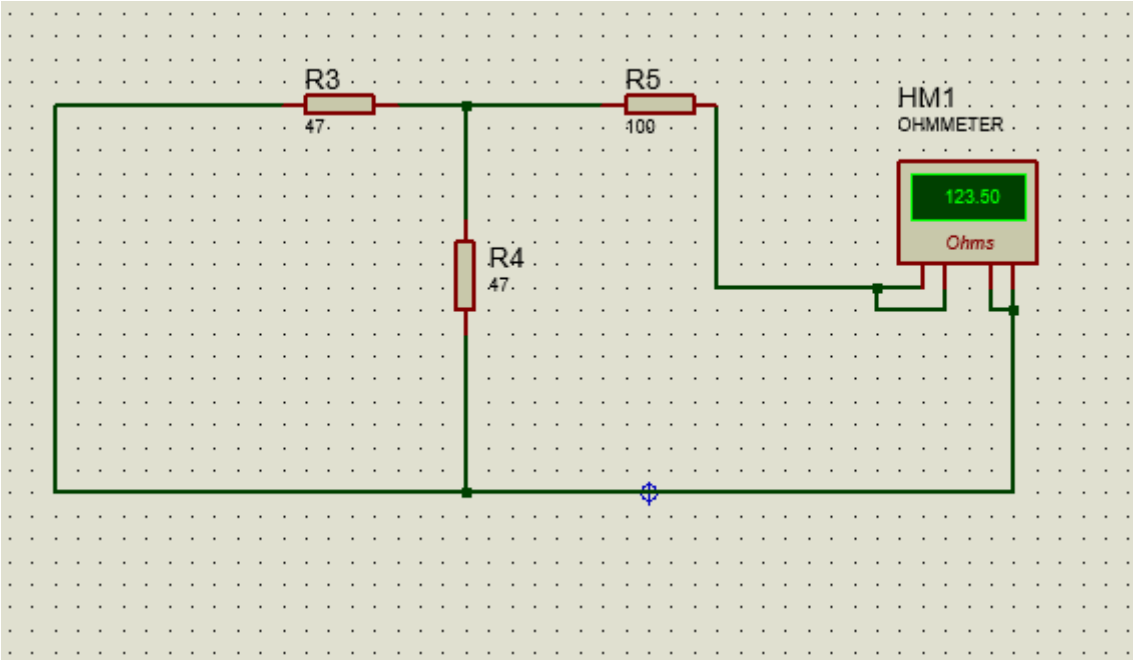
$$P_L = \left(\frac{10}{12k}\right)^2 (6k) = \frac{25}{6} mW$$

Active Learning

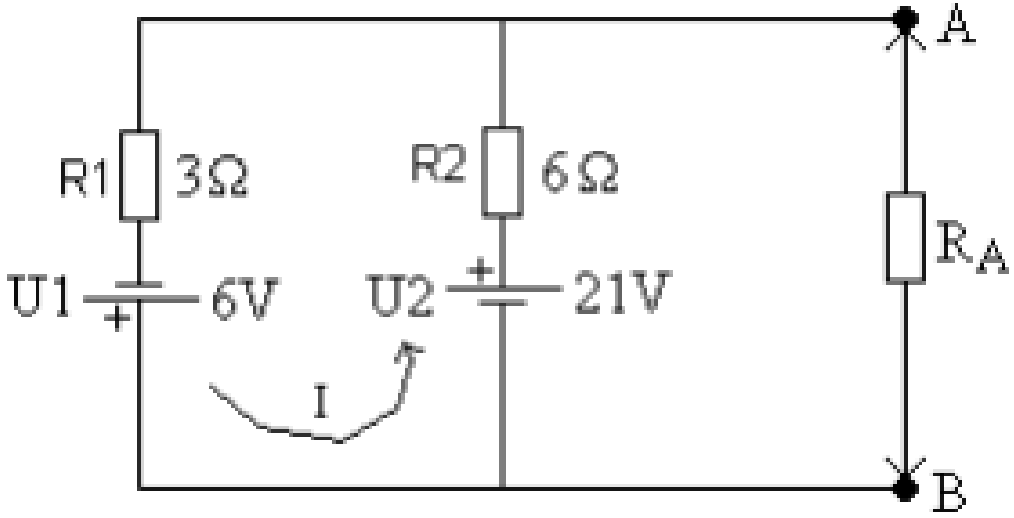
1. Şekildeki devre için Norton Akımı IN, Norton direnci RN değerlerini hesaplayınız.

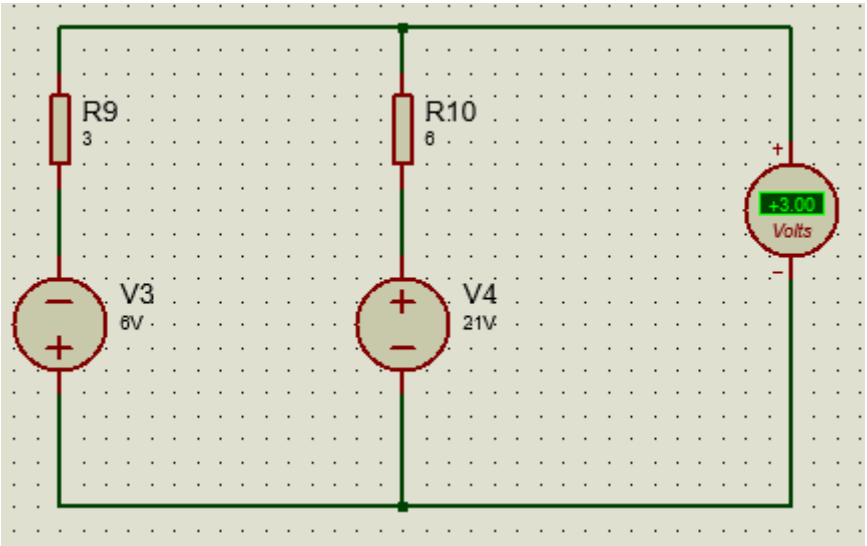


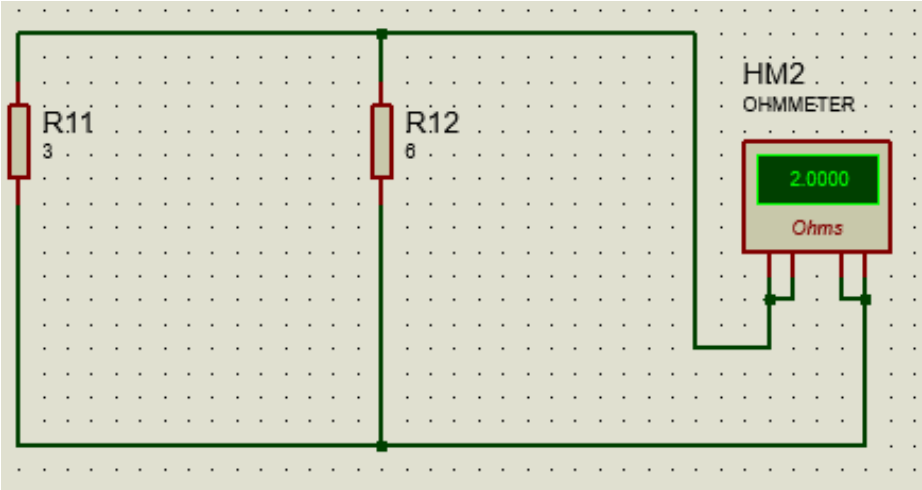

R3
47
R5
100
V1
5V
R4
47
+
Amps
+0.02
-

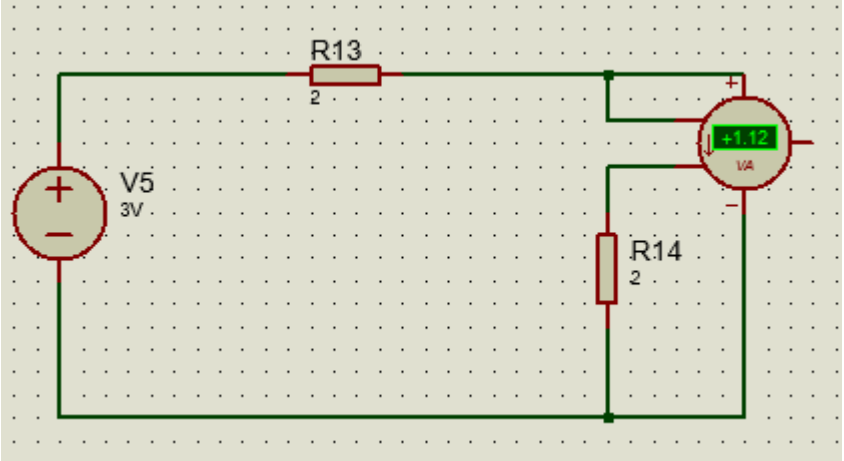

R3
47
R5
100
HM1
OHMMETER
123.50
Ohms
R4
47

2. Şekildeki devrede R_A direncinin maksimum güç çekebilmesi için değeri ne olmalıdır. Thevenin eşdeğer devresi yardımıyla R_A direncinin çektiği maksimum gücü hesaplayınız.

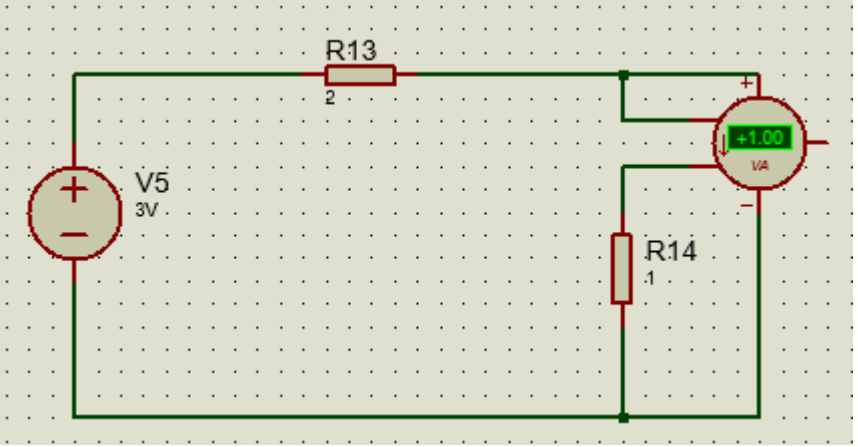



R9
3
R10
6
+
+3.00
Volts
-
V3
6V
V4
21V

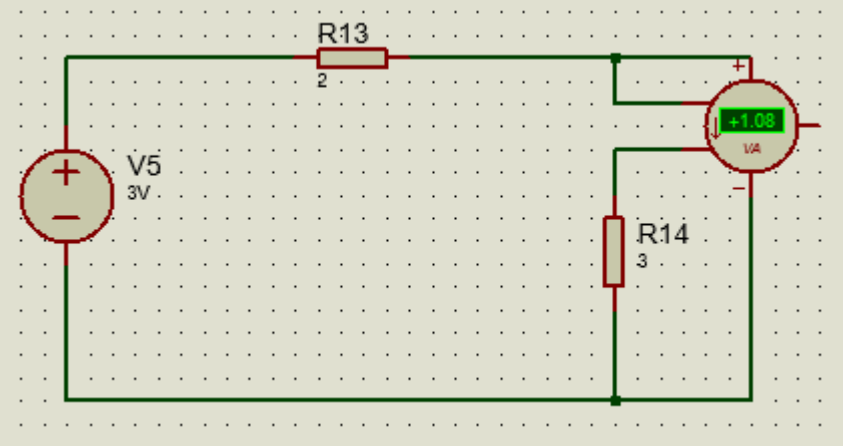

R11
3
R12
6
HM2
OHMMETER
2.0000
Ohms



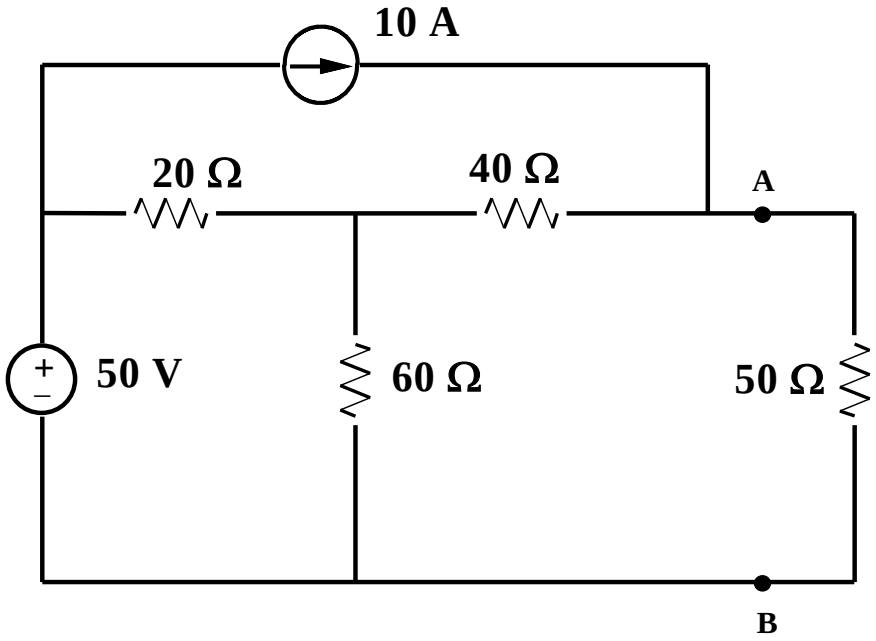
- Düşük Yük Direnci



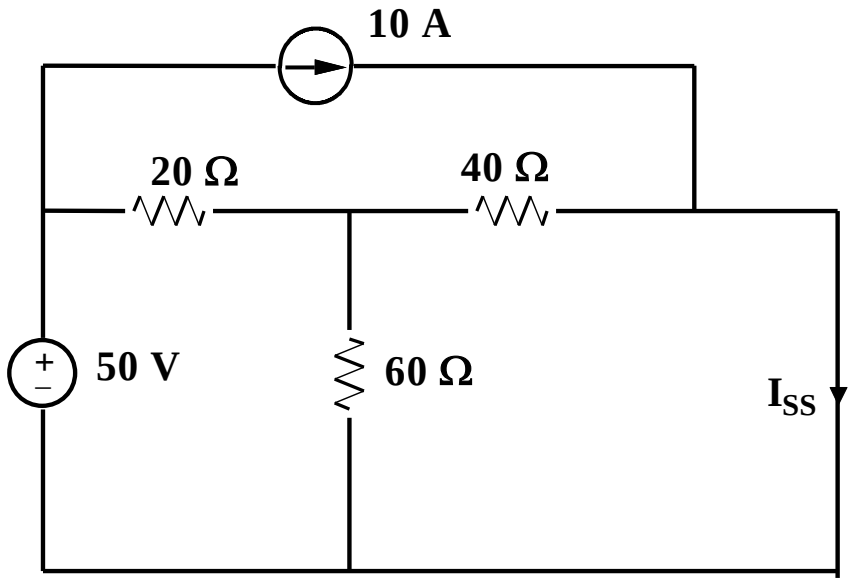
Yüksek Yük Direnç



- Aşağıdaki devrede A-B noktaları arasındaki 50 ohmluk dirençten geçen akımı Norton teoremine göre bulunuz



- In hesaplayalım

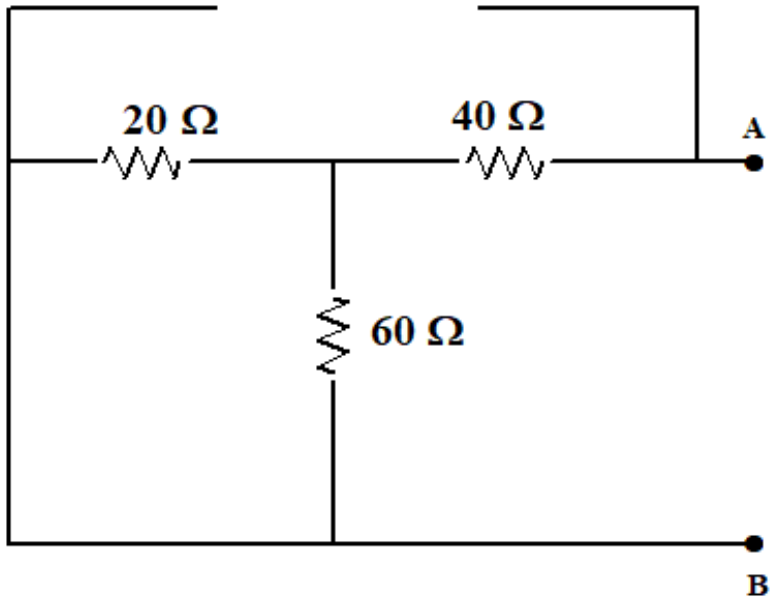


- Düğüm analizi ile kolayca bulunabilir

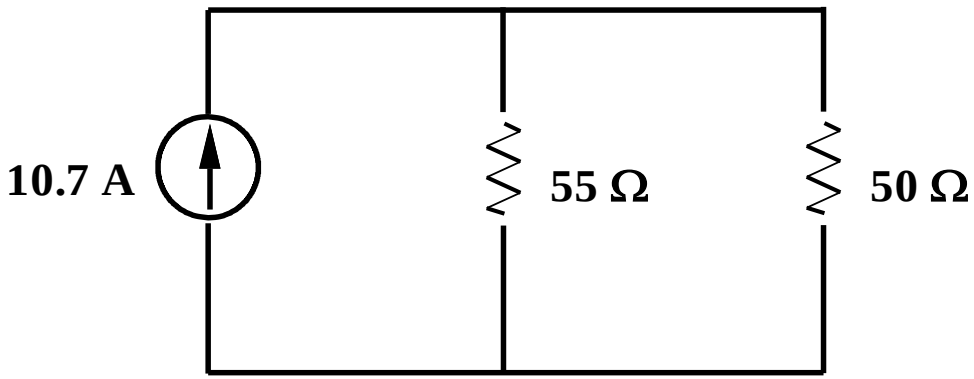
$$I_{SS}=10.7\ A$$

- Rn bulalım

- Rn=55 ohm

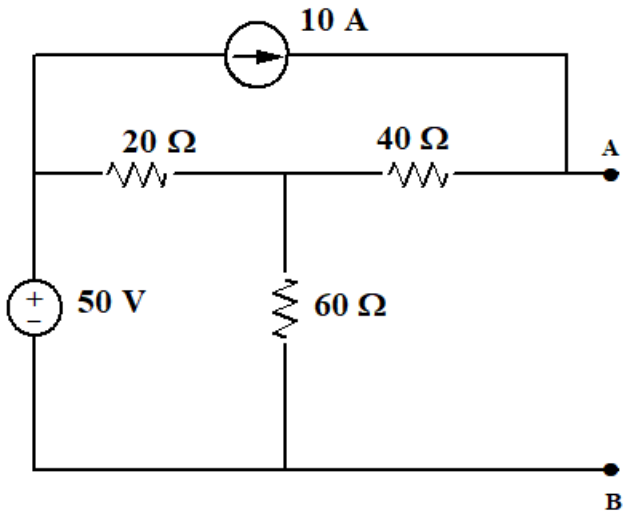


- Eşdevreyi çizelim

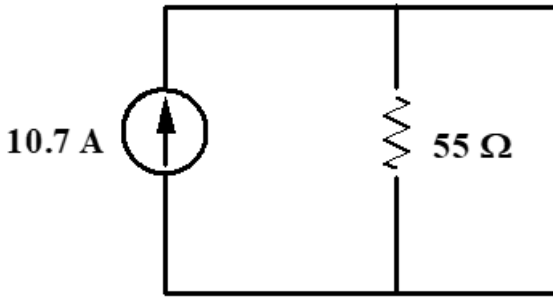


- Buradan 50 ohm luk dirençten geçen akım I=4.67A olur

- 4- bir önceki soruda A-B uçları arasına maksimum güç aktarabilmek için kaç ohmuk yük bağlamak gerekir



- Norton eşdeğerinden thevenin eşdeğerini elde edelim



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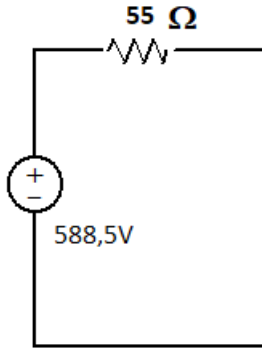
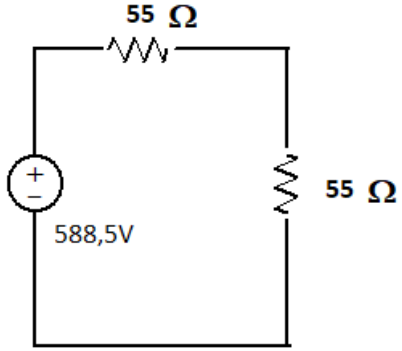
maksimum güç için: